

Guided-Wave Suppression in Uniaxial Anisotropic Metamaterial Slab for Radiation Pattern Control of Covered Dipoles

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Abstract—A dipole positioned in parallel close to a dielectric cover excites strong guided waves, typically distorting its radiation patterns. This study presents conditions for suppressing guided waves in a uniaxial anisotropic slab and then proposes a uniaxial anisotropic meta-cover to improve the un-roundness of distorted radiation patterns of off-center positioned dipoles. The proposed metamaterial cover (meta-cover in short) of periodic metal mesh unit cells exhibits uniaxial anisotropic properties with electric Drude-model dispersion, suitable for wideband guided-wave suppression. For comparison, the proposed meta-cover and conventional homogeneous dielectric are designed and used to closely cover off-center omnidirectional horizontally polarized four-dipole rings (FDRs). The study theoretically and experimentally verifies that by suppressing the guided waves in the meta-cover, the FDRs achieve an improvement in the un-roundness of the radiation pattern from 17.6 to 5.2 dB over the frequency range from 5.0 to 6.0 GHz, so that coverage efficiency increases from 42.4% to 84.7%. The derived conditions to suppress the guided waves in a meta-cover benefit mitigating cover influences on the radiation performance of antennas in modern wireless systems.

Index Terms—Anisotropic metamaterial, cover, dipole, radiation pattern.

I. INTRODUCTION

MODERN antenna systems impose stringent requirements on radiation patterns to achieve high capacity, reliability, and data rates. Specifically, systems such as wireless local-area network (WLAN) access points (APs) and micro base stations necessitate omnidirectional radiation patterns for each antenna within the multiple-input multipole-output (MIMO) system. This ensures wide coverage, eliminates dead zones, and maximizes coverage efficiency, as indicated in Fig. 1.

Antenna radiation patterns are sensitive to nearby environments. Restoring radiation pattern distortion has been more and more discussed. For vertically polarized antennas on finite ground, the radiation distortion is mainly caused by irregular ground planes. Approaches have been proposed such

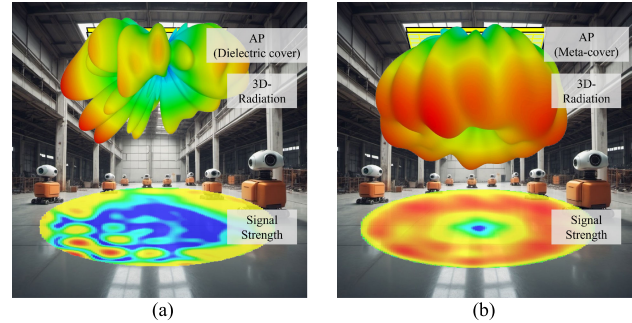


Fig. 1. Diagrams of APs mounted on the ceiling with (a) dielectric cover and (b) meta-cover.

as circular ring [1], metasurface [2], circular electromagnetic bandgap (EBG) [3], and henge-like metaline ring for improving radiation patterns [4]. Besides antenna elements and ground planes, antenna covers also stand as an essential component within nearly all antenna systems for providing crucial protection to antennas and surrounding electronics against adversarial external elements [5]. The conventional dielectric covers have been found to influence the radiation patterns since 1996 [6] and reintroduced in 2021 [7]. Till now, beyond protection, antenna covers or radomes are also utilized to incorporate diverse functionalities that enhance the performance of the antenna system, such as frequency selective surface for spatial filters [8], [9], [10], [11], [12], low radar cross section [8], out-of-band absorber [10], extended scanning range of a phased-array antenna [13], [14], [15], and wide-angle polarization-insensitive electromagnetic windows for high transmission [16], [17], [18].

Among these applications, plane wave incidence is assumed for analysis, which indicates the covers must be mounted a distance from the antennas to satisfy the far-field conditions. However, for more widely applied scenarios, such as base stations and APs, the covers are placed close to the antennas [19], [20], [21], where the plane wave incidence assumption is not satisfied, and the covers influence the antenna's matching, coupling, and radiation patterns. Most of the works [19], [20], [21] codesign the antennas and covers for well-matching.

Recently, single-negative metamaterial (SNM) has been proposed for suppressing the coupling between antennas caused by guided waves in the covers [22], [23], [24]. While it is not rigorous to assume the wave traveling in a free space and

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analysis with wavenumber in a space full of materials, a more accurate model for the cover is a slab waveguide.

Alternatively, metamaterials (MTMs) as artificial structures with unique electromagnetic (EM) properties that have yet to be found in nature [25], [26], achieve flexible permittivity and permeability tensors to manipulate EM waves with high degrees of freedom. For instance, MTMs have been widely applied as EM bandgaps for decoupling [27], dispersion-engineered for wideband antennas [28], zero-index materials for directivity enhancement [29], [30], nonlinear metasurfaces [31], magnon effects for optomagnonic cavities [32], and so on.

Anisotropic MTMs feature flexibility in manipulating EM waves. Their fascinating applications include transformation optics [33], cloaking [34], [35], polarization conversion [36], [37], and so on. The anisotropic MTMs have been theoretically studied in metallic waveguides [38], [39], [40], [41], [42], grounded slab waveguides [43], [44], and dielectric slab waveguides [45], [46], [47], [48], [49], [50], [51], [52], [53], [54]. The wave equations and dispersion relations have been derived for inhomogeneous media [38], layered structures [45], [46], [47], arbitrary directions of optical axis [45], [46], and asymmetric waveguides [48]. These papers share one important feature that the waveguiding media shows only anisotropy of permittivity, while the relative permeability remains unity. The discussions of both permittivity and permeability in the forms of tensors emerged from the realization of MTMs with negative permeability [55] and negative refraction index [56].

The study of exotic properties of waves in double negative (DNG) slab waveguides [51], [52] reveals peculiar properties, such as the absence of the fundamental modes, mode double degeneracy, and sign-varying energy flux. He and Eleftheriades [17] show the guidance conditions and mode properties of transverse electric (TE) waves in a grounded slab with uniaxial anisotropic left-handed material. Then, the characteristics of EM waves in a slab waveguide of uniaxial cold plasma media are presented in [53]. The necessary and sufficient conditions for independent TE and transverse magnetic (TM) modes in a rectangular metallic waveguide with constitutive parameters in double-positive Hermitian matrices are rigorously derived in [54], where a uniaxial material of independent modes with the direction of an optical axis parallel to the normal direction of the slab waveguide is a special case. Such a phenomenon agrees with one in [43] and will be applied to the following derivations for simplicity.

In this article, the waveguiding conditions of both TE and TM modes in a lossless uniaxial anisotropic slab waveguide with optical axial normal to the thin slab are derived with arbitrary permittivity and permeability. Based on the cut-off condition, a meta-cover is designed to improve the radiation pattern of the off-center dipole ring antennas by suppressing the surface waves using the designed uniaxial anisotropic slab.

II. CUT-OFF CONDITIONS OF UNIAXIAL METAMATERIAL SLABS

This section presents a comprehensive analysis of the cutoff conditions governing the propagation of EM waves in uniaxial metamaterial slabs. The derivation commences without

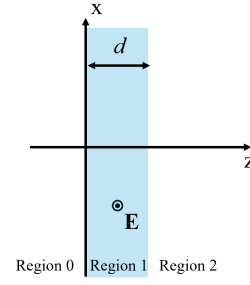


Fig. 2. Schematic of a slab waveguide with thickness, d .

imposing any prior constraints on the values of permittivity and permeability, thereby ensuring the generality of the conclusions drawn.

Considering a thin slab with a thickness of d ($d \ll \lambda_0$, λ_0 is the operating wavelength in free space) as shown in Fig. 2. Regions 0 and 2 are air while Region 1 is filled with lossless uniaxial anisotropic media with the relative permittivity and permeability as follows:

$$\bar{\bar{\epsilon}}_{r1} = \mathbf{a}_x \mathbf{a}_x \epsilon_t + \mathbf{a}_y \mathbf{a}_y \epsilon_t + \mathbf{a}_z \mathbf{a}_z \epsilon_z \quad (1)$$

$$\bar{\bar{\mu}}_{r1} = \mathbf{a}_x \mathbf{a}_x \mu_t + \mathbf{a}_y \mathbf{a}_y \mu_t + \mathbf{a}_z \mathbf{a}_z \mu_z \quad (2)$$

where $\mathbf{a}_x$, $\mathbf{a}_y$, and $\mathbf{a}_z$ are unit vectors along x -, y -, and z -axis, respectively, ϵ_t and μ_t are the transverse relative permittivity and permeability along the x - and y -axis, and ϵ_z and μ_z are the normal relative permittivity and permeability along the z -axis.

In the three regions, a TE wave is propagating along the x -axis, written as

$$\mathbf{E}_0 = \mathbf{a}_y E_0^- e^{\alpha_{0z} z} e^{-jk_x x} \quad (3)$$

$$\mathbf{E}_1 = \mathbf{a}_y (E_1^+ e^{jk_{1z} z} + E_1^- e^{-jk_{1z} z}) e^{-jk_x x} \quad (4)$$

$$\mathbf{E}_2 = \mathbf{a}_y E_2^+ e^{-\alpha_{0z} z} e^{-jk_x x} \quad (5)$$

The magnetic fields are

$$\mathbf{H}_0 = \frac{1}{j\omega\mu_0} (\mathbf{a}_z jk_x - \mathbf{a}_x \alpha_{0z}) E_0^- e^{\alpha_{0z} z} e^{-jk_x x} \quad (6)$$

$$\begin{aligned} \mathbf{H}_1 = & \frac{1}{\omega} \left(\mathbf{a}_z \frac{k_x}{\mu_z \mu_0} - \mathbf{a}_x \frac{k_{1z}}{\mu_t \mu_0} \right) E_1^+ e^{jk_{1z} z} e^{-jk_x x} \\ & + \frac{1}{\omega} \left(\mathbf{a}_z \frac{k_x}{\mu_z \mu_0} + \mathbf{a}_x \frac{k_{1z}}{\mu_t \mu_0} \right) E_1^- e^{-jk_{1z} z} e^{-jk_x x} \end{aligned} \quad (7)$$

$$\mathbf{H}_2 = \frac{1}{j\omega\mu_0} (\mathbf{a}_z jk_x + \mathbf{a}_x \alpha_{0z}) E_2^+ e^{-\alpha_{0z} z} e^{-jk_x x} \quad (8)$$

where k_x is the wavenumber along the x -axis, while k_{1z} is the wavenumber in Region 1 along the z -axis. Matching the boundary conditions on both the interfaces of the slab, the guidance conditions for the real part of transverse wavenumber k_{1z} in even modes and odd modes are

$$\text{Even mode: } \alpha_{0z} d = \frac{1}{\mu_t} (k_{1z} d) \tan\left(\frac{k_{1z} d}{2}\right) \quad (9)$$

$$\text{Odd mode: } \alpha_{0z} d = -\frac{1}{\mu_t} (k_{1z} d) \cot\left(\frac{k_{1z} d}{2}\right). \quad (10)$$

Similarly, for the imaginary part of transverse wavenumber $k_{1z} = j\alpha_{1z}$

$$\text{Even mode: } \alpha_{0z} d = -\frac{1}{\mu_t} (\alpha_{1z} d) \tanh\left(\frac{\alpha_{1z} d}{2}\right) \quad (11)$$

$$\text{Odd mode: } \alpha_{0z} d = -\frac{1}{\mu_t} (\alpha_{1z} d) \coth\left(\frac{\alpha_{1z} d}{2}\right). \quad (12)$$

TABLE I
SUMMARY OF CUT OFF CONDITIONS OF TE WAVES IN UNIAXIAL ANISOTROPIC SLAB WAVEGUIDE

		k_{1z} is real		k_{1z} is imaginary		Wave behaviors
		Even	Odd	Even	Odd	
$T_1 > 0$	$T_2 > 0$	$\mu_t > 0$	Propagate	Cut off ¹	Cut off	Propagate
		$\mu_t < 0$	Propagate	Propagate	Cut off	Propagate
	$T_2 < 0$	$\mu_t > 0$	Propagate	Propagate	Cut off	Propagate
		$\mu_t < 0$	Propagate	Propagate	Cut off ²	Propagate
$T_1 < 0$	$T_2 > 0$	$\mu_t > 0$	Cut off	Cut off	Cut off	Cut off
		$\mu_t < 0$	Cut off	Cut off ³	Cut off ⁴	Cut off ⁵
	$T_2 < 0$	$\mu_t > 0$	Propagate	Propagate	Cut off	Propagate
		$\mu_t < 0$	Propagate	Propagate	Cut off ²	Propagate

¹: Cut off when: $\sqrt{T_1/T_2} < \lambda/(2d)$; ²: Cut off when: $\varepsilon_t T_1 k_0^2 d^3 > -2$; ³: Cut off when: $\varepsilon_t > 1/\mu_z - \mu_z/(k_0 d)^2$; ⁴: Cut off when: $0 < \mu_z \mu_t < 1$; ⁵: The cut off condition is satisfied when both conditions 3 and 4 are satisfied

Besides, the dispersion relation in Region 0 is

$$(k_x d)^2 - (\alpha_{0z} d)^2 = (k_0 d)^2. \quad (13)$$

The dispersion relations in Region 1 with a real or imaginary part of transverse wavenumber are, respectively

$$(k_x d)^2 + \frac{\mu_z}{\mu_t} (k_{1z} d)^2 = \varepsilon_t \mu_z (k_0 d)^2 \quad (14)$$

$$(k_x d)^2 - \frac{\mu_z}{\mu_t} (\alpha_{1z} d)^2 = \varepsilon_t \mu_z (k_0 d)^2. \quad (15)$$

Eliminating $k_x d$ by combining (13) with (14) and (15), we obtain

$$(\alpha_{0z} d)^2 + \frac{\mu_z}{\mu_t} (k_{1z} d)^2 = (\varepsilon_t \mu_z - 1) (k_0 d)^2 \quad (16)$$

$$(\alpha_{0z} d)^2 - \frac{\mu_z}{\mu_t} (\alpha_{1z} d)^2 = (\varepsilon_t \mu_z - 1) (k_0 d)^2. \quad (17)$$

Combining the dispersion relations (16) and (17) with the boundary conditions (9)–(12), equations containing unknowns $k_{1z} d$ or $\alpha_{1z} d$ in different cases are obtained. To find the cut-off conditions, diagram methods are applied: the guided wave can propagate in the anisotropic slab waveguide if the corresponding equations have a cross point. The wave behaviors in the slab waveguide are listed in Table I, where $T_1 \equiv (\varepsilon_t \mu_z - 1)$ and $T_2 \equiv \mu_z / \mu_t$.

The properties of the wave in the last column represent the union set for cut-off conditions. For example, ‘cut-off’ in the column means this kind of uniaxial anisotropic meta-cover does not support guided TE waves, regardless of whether the normal wavenumber is real or imaginary, nor the propagation mode is even or odd. Thus, the cut-off conditions for all kinds and all orders of guided TE waves in a uniaxial anisotropic slab waveguide can be satisfied in either of the two conditions: In Condition 1.1 $T_1 < 0$, $T_2 > 0$ and $\mu_t > 0$. In Condition 1.2, the cut-off depends on $k_0 d$: $T_1 < 0$, $T_2 > 0$, $\mu_t < 0$, $\varepsilon_t > (1/\mu_z) - (\mu_z/(k_0 d)^2)$, and $0 < \mu_z \mu_t < 1$. The conditions can be formulated in terms of material properties as follows:

Condition 1.1: $\varepsilon_t \mu_z < 1$, $\mu_t > 0$, $\mu_z > 0$.

Condition 1.2: $\varepsilon_t \mu_z < 1 - (\mu_z/k_0 d)^2$, $\mu_t < 0$, $\mu_z < 0$, $\mu_z \mu_t < 1$.

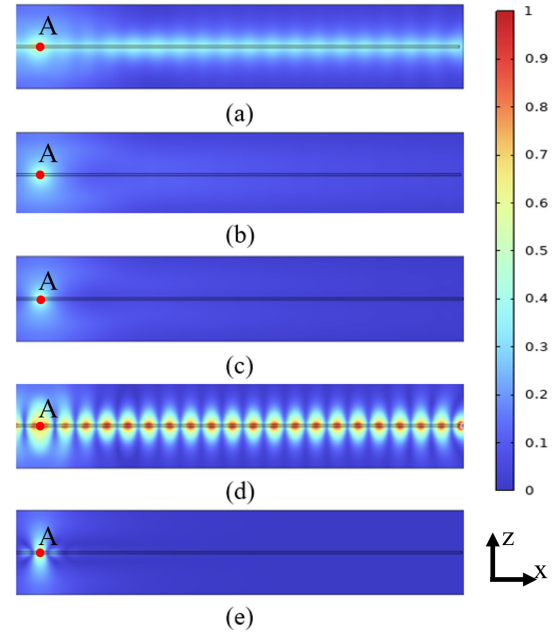


Fig. 3. Magnitude of the E -field distribution along the slab waveguide. The right red point indicates the exciting source. (a) $\varepsilon_t = 4$, $\mu_t = \mu_z = 1$. (b) $\varepsilon_t = 0.2$, $\mu_t = \mu_z = 1$. (c) $\varepsilon_t = -2$, $\mu_t = \mu_z = 1$. (d) $\varepsilon_t = 4$, $\mu_t = \mu_z = -2$. (e) $\varepsilon_t = 4$, $\mu_t = -1$, $\mu_z = -0.3$.

For TE waves, satisfying any of these two conditions can suppress any order of guided waves in a uniaxial anisotropic slab waveguide.

The cut-off of TM waves can be obtained by duality.

Condition 2.1: $\varepsilon_z \mu_t < 1$, $\varepsilon_t > 0$, $\varepsilon_z > 0$.

Condition 2.2: $\varepsilon_z \mu_t < 1 - (\varepsilon_z/k_0 d)^2$, $\varepsilon_t < 0$, $\varepsilon_z < 0$, $\varepsilon_z \varepsilon_t < 1$.

The findings are verified by filling a slab waveguide with materials in simulations as the central thin slab shown in Fig. 3. The slab thickness is $1/20 \lambda_0$, where λ_0 is the wavelength in free space at 5 GHz. One port is excited at point A with field $E_y = 1$ V/m for an excited TE wave, for example.

In Fig. 3(a), the isotropic dielectric with $\varepsilon_r = 4$ and $\mu_r = 1$ as a reference is used as the filled materials. From the magnitude of the E -field plotted in Fig. 4, it is found that

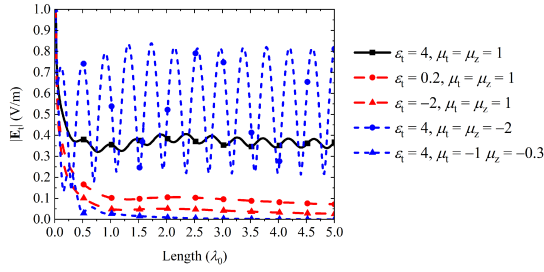


Fig. 4. Magnitude of E -field along the center of the slab waveguide filling with different materials.

the standing guided waves are well-excited along the slab waveguides due to the reflection of the waves at the ends of the slab waveguides. As shown in both Figs. 3 and 4, the guided waves are well excited in the reference slab waveguide so that EM energy is almost preserved in the waveguide.

Next, to satisfy Condition 1.1 in (17) for suppressing the guided waves of TE mode, two examples are simulated with $\varepsilon_t = 0.2$ and $\varepsilon_t = -2$ while keeping the $\mu_t = \mu_z = 1$. The guided waves propagating along the slab waveguide are well suppressed as shown in Fig. 3(b) and (c). Besides, the magnitude of E_t is reduced to lower than 0.1 V/m after propagating over a one-wavelength distance in the slab, at least 70% lower compared with the reference case on average. Therefore, satisfying Condition 1.1 is sufficient to suppress the TE waves in a slab waveguide. However, the condition is less restrictive than the epsilon-negative (ENG) materials.

Two more examples are presented to verify Condition 2.1 as shown in Fig. 3(d) and (e). In Fig. 3(d), the μ -negative material is set as $\varepsilon_t = 4$ and $\mu_t = \mu_z = -2$. A well-excited guided wave is propagating in the slab with a magnitude even larger than the reference in Fig. 3(a). The wave propagates on the interface between the air and the slab and decays in the normal direction of the slab, similar to surface plasmon on the interface between an ENG material and air. This phenomenon shows that a slab with a single negative material can also guide EM waves although the wavenumber is purely imaginary. However, the guided waves can be suppressed if satisfying Condition 2.1 as shown in Fig. 3(e), where the $\varepsilon_t = 4$, $\mu_t = -1$, and $\mu_z = -0.3$.

In Fig. 4, the magnitude decreases exponentially when either of the conditions is satisfied for TE waves. The conclusions for TM waves can be proved by duality and thus will be omitted here for concise.

In short, the conditions for suppressing TE or TM-guided waves in a uniaxial anisotropic slab waveguide are derived and verified. These conditions provide sufficient and effective guidance for guided wave suppression.

III. META-COVER ANALYSIS AND DESIGN

The derived conditions are applied to a meta-cover, as an example, to suppress guided waves and improve radiation patterns of dipoles with a dielectric cover. Fig. 5 shows a four-dipole ring (FDR) formed by four curved dipoles fed by one feeding port and in phase for horizontally polarized omnidirectional radiation. Then, four FDRs are positioned above a square ground and covered by a dielectric slab to compose a four-port MIMO antenna system. The antenna system operates

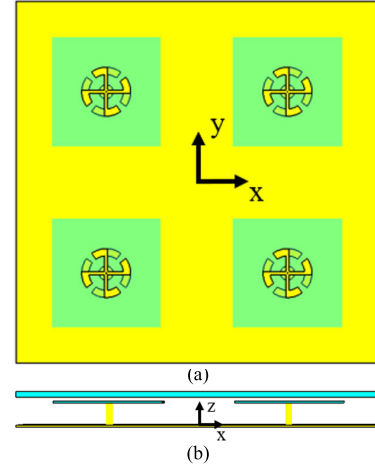


Fig. 5. Configuration of an MIMO system consisting of four horizontally polarized FDRs sharing a finite ground plane and cover. (a) Top view. (b) Side view.

in a frequency range from 5 to 6 GHz. The ground and cover are 200×200 mm or $3.7\lambda_0 \times 3.7\lambda_0$ at 5.5 GHz. The FDRs are located at $(x, y) = (50, 50)$ mm, $(-50, 50)$ mm, $(50, -50)$ mm, and $(-50, -50)$ mm, respectively, where the origin of the coordinate system is located at the center of the ground. The cover is 3 mm thick, parallel to the ground, and 2 mm above the dipoles.

A. Influence of Dielectric Cover on Off-Center FDRs

Because of the limited size of the antenna system with multiple antennas sharing one environment, antennas must be mounted off the center of the ground to reduce mutual coupling effects. Studies have shown that off-center monopole antennas above a finite ground suffer from distorted radiation patterns [1], [2], [3], [4]. FDRs, however, pose the same challenge but with different physical essences due to their horizontal polarization.

As shown in Fig. 6, five cases are discussed for comparison as follows:

Case 1: FDR above an infinite ground without cover as a reference.

Case 2: Off-center FDR above a finite ground without a cover.

Case 3: Off-center FDR above an infinite ground with a cover.

Case 4: Off-center FDR above a finite ground with a cover.

Case 5: Four off-center FDRs forming an MIMO system above a finite ground with a cover.

The covers are isotropic dielectric materials ($\varepsilon_t = 4$). The corresponding electric field distribution in the cutting plane of $y = 50$ mm is shown in Fig. 6(f)–(j). Fig. 6(f) and (h) only plots the fields in the upper space.

Different from monopoles, the diffraction at the edges of the finite ground is almost nonexistent in Case 2, as shown in Fig. 6(g) because the FDR operates as a vertically polarized magnetic source, which cannot effectively induce tangential H -field on the ground, correspondingly limiting surface currents traveling along the ground. Therefore, weak diffraction at the edges of the ground keeps the radiation patterns omnidirectional and less distorted.

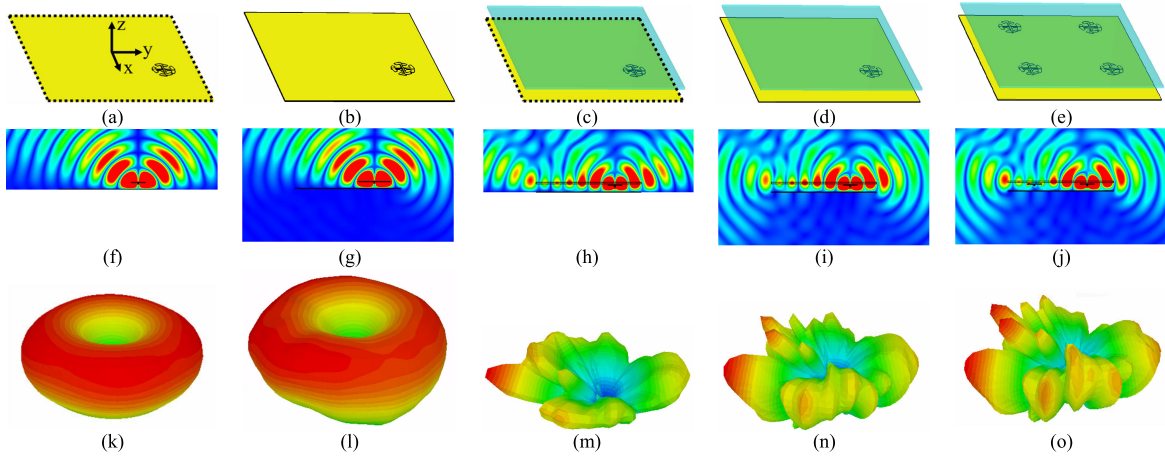


Fig. 6. (a)–(e) Configurations of Cases 1–5 (dotted boundaries indicate infinite ground). (f)–(j) E -field distribution of Cases 1–5 in $y = 50$ cutting plane. (k)–(o) 3-D normalized radiation patterns of Cases 1–5.

The introduction of a cover, however, usually generates strong guided waves in the cover as shown in Fig. 6(h)–(j), where the gap between the cover and antennas is only $0.037\lambda_0$ and the FDRs are parallel to the cover surface. Since the off-center location of the FDR, the radiation of the guided waves at the edges near and far from the dipoles is different. The overall antenna beam is more directed to the edge of the cover far from the dipole and points more to endfire. Therefore, the radiation is asymmetric and heavily distorted.

Moreover, comparing the single FDR in Fig. 6(d) with the four FDRs in Fig. 6(e), the field distribution and radiation patterns are almost the same. This is because of the relatively far distance ($1.8\lambda_0$) and weak coupling between different FDRs. Therefore, the guided waves in the dielectric cover severely distort the radiation patterns of a finite-sized horizontally polarized MIMO antenna system, while the finite ground and the nearby antennas have limited effects.

B. Transmission Properties of Meta-Cover

Besides guided waves, the transmission properties of meta-cover are crucial for alleviating the distortion of the radiation patterns of antennas. Four cases, where a point source is placed 2 mm below an infinite anisotropic meta-cover, are simulated for studying the transmission properties of the meta-cover, as shown in Fig. 7. In all simulations, ε_z is set to 4, and μ_r is set to 1. The point source is y -polarized and radiates isotropically in the xz plane. The thickness of the meta-cover is 3 mm.

Fig. 7(a) shows that the wave radiated from the source is mostly reflected by the meta-cover but does not excite guided waves in the slab because the material properties of the meta-cover satisfy Condition 1.1, namely, $\varepsilon_t \mu_z < 1$, $\mu_t > 0$, $\mu_z > 0$. As the transverse permittivity of the slab $|\varepsilon_t|$ equals the unit, the meta-cover becomes fully transparent as shown in Fig. 7(b) and (c). The only difference between them is that there are no guided waves along the slab in Fig. 7(b), that is the wave energy toward the $\pm x$ -axis tends to be zero. In Fig. 7(d), part of the energy penetrates the meta-cover while the others transfer to guided waves along the slab as $\varepsilon_t = 10$.

A qualitative analysis of the transmitted E -field is plotted in Fig. 8. Testing probes are placed at two positions one

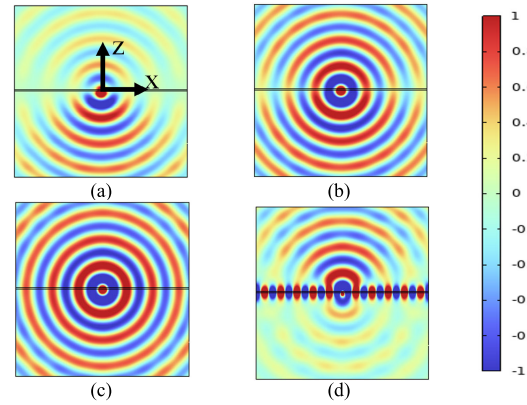


Fig. 7. Electric field distribution of a y -polarized isotropic point source below the meta-cover. (a) $\varepsilon_t = -10$. (b) $\varepsilon_t = -1$. (c) $\varepsilon_t = 1$. (d) $\varepsilon_t = 10$.

wavelength above the slab at $\theta = 0^\circ$ and $\theta = 60^\circ$, as shown in the insertion. The magnitude of the E -field ($|\mathbf{E}_{\text{meta}}|$) at the testing positions is normalized by the magnitude of the E -field at the same location without the meta-cover ($|\mathbf{E}_{\text{air}}|$) for comparing the influence of the meta-cover on the radiation of the closely placed point source at two directions. The light blue and the dark blue area stand for the transmitted E -field at least reaches 80% at Positions 1 and 2, respectively. The results show that: 1) transmission decreases when negative ε_t keeps decreasing, which is caused by reflection; 2) large positive ε_t decreases the transmission because the slab transfers the energy of EM waves in free space to guided waves along the slab; and 3) transmission is acceptable at different angles with ε_t in the range from -1.4 to 5.7 , suitable for meta-cover design with acceptable transparency.

C. Meta-Cover Unit Cell and Meta-Cover Design

Being aware of the mechanism of the radiation distortion of the FDRs, a unit cell for meta-cover is designed to satisfy the required material properties as derived and concluded in Section II. Since FDR is TE polarized, the meta-cover is designed to suppress the TE-guided waves in the following.

From the previous study, it is found that satisfying Condition 1.1 provides more potential for a simple design with easily achievable properties of material although satisfying

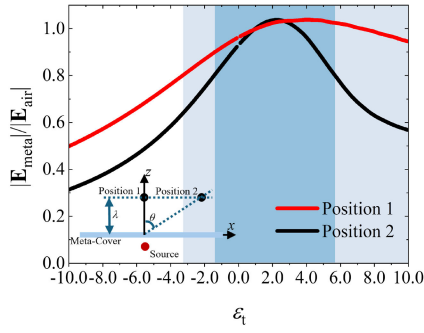


Fig. 8. Normalized magnitude of $\mathbf{E}$ -field at Positions 1 and 2 with varying ϵ_t .

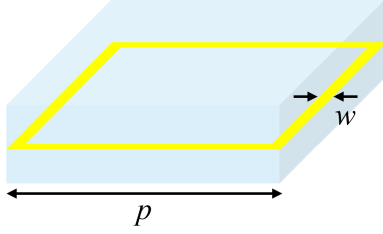


Fig. 9. Drude-model unit cell geometry ($p = 11.5$ mm and $w = 0.1$ mm).

either Condition 1.1 or 1.2 is sufficient. Thus, the material of the meta-cover should satisfy $-1.4 < \epsilon_t < 5.7$, $\epsilon_t \mu_z < 1$, $\mu_t > 0$, and $\mu_z > 0$, where $\mu_z \approx \mu_t \approx 1$.

Generally, materials with negative real parts of constituting parameters can be categorized into the Lorentz model and the Drude model [57]. The near-zero dispersive properties of these two models are analyzed in Appendix. It is concluded that Drude model unit cells are less dispersive at plasma frequency, where $\text{Real}(\epsilon) = 0$. Therefore, a Drude-model unit cell is designed.

Physically, Drude model requires free electrons between unit cells, as the wire medium proposed in [58]. Inspired by this, a square mesh unit cell is proposed as shown in Fig. 9, where the metal mesh is outermost and connected to the nearby cells. The equivalent material properties of ϵ_t , ϵ_z , μ_t , and μ_z in Fig. 10 are extracted from reflection and transmission coefficients calculated by illuminating the unit cell with varying incident angles of plane waves [59]. It can be found that the properties of the unit cells satisfy the conditions of TE-guided wave suppression and high transmission from 5.0 to 6.2 GHz, and in particular, imaginary parts of all material properties are nearly zero, namely, achieving ignorable insertion loss. Then, the meta-cover is formed with the uniaxial anisotropic unit cells.

D. FDR Design

The FDRs are designed as horizontally polarized omnidirectional antennas, as shown in Fig. 11. The FDRs are all fed in phase through a coaxial line at the center of the antennas, where the inner conductor is connected to the upper-layer metal, and the outer conductor is connected to the lower-layer metal. The radiating dipole arms are fed by parallel double strips, where the meandering lines are for impedance matching.

Additionally, Section II provides the theoretical analysis and conclusions regarding guided waves, assuming invariance along the y -axis. Meanwhile, because of the in-phase feeding

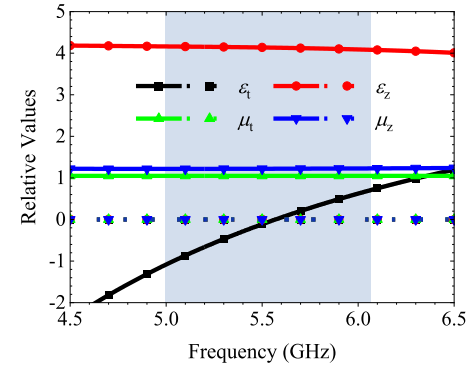


Fig. 10. Equivalent material properties of a unit cell in Fig. 9. Solid and dotted lines are real and image parts, respectively. The light blue area stands for values that satisfy the meta-cover conditions to restore the radiation pattern.

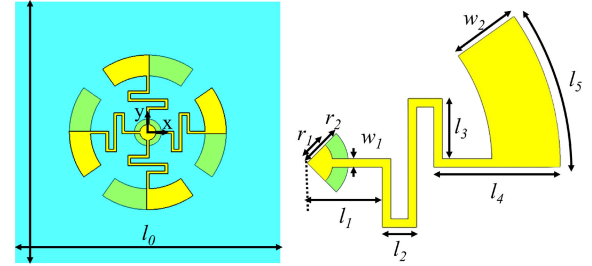


Fig. 11. Top view and detailed geometries of the FDR beneath the meta-cover ($l_0 = 50$, $l_1 = 4.4$, $l_2 = 2$, $l_3 = 3.5$, $l_4 = 7.4$, $l_5 = 9.4$, $w_1 = 0.5$, $w_2 = 4$, $r_1 = 1.5$, $r_2 = 1.5$, $r_2 = 2.1$, unit mm).

and symmetry of the FDRs, the radiated fields are almost invariant along the φ -axis in a cylindrical coordinate, which is centered at the geometry center of the FDR with the z -axis perpendicular to it. Therefore, the guided-wave suppression conditions can be applied in the practical FDRs being discussed later.

E. MIMO Antenna With Meta-Cover

The configuration of the meta-cover consisting of the proposed unit cells and antennas is shown in Fig. 5. The FDR is 13 mm away from the ground, and the cover is 2 mm above the FDRs.

The magnitude of the $\mathbf{E}$ -field and corresponding 3-D radiation patterns are plotted in Fig. 12(a) and (b), with the ϵ_t set to 4 and -0.1 at 5.5 GHz, respectively (where $\epsilon_z = 4$, $\mu_z = \mu_t = 1$ in all cases). As studied previously, conventional dielectric cover couples strong guided waves, as shown in Fig. 12(a), and the corresponding radiation pattern is severely distorted. By applying the uniaxial anisotropic meta-cover, whose material properties satisfy the guided wave suppressing and high-transmission conditions, it can be found that the guided waves are well suppressed, and the radiation pattern is recovered to omnidirectional, as shown in Fig. 12(b). Then, the material of the meta-cover in Fig. 12(b) is realized by the mesh unit cells designed in Section III-C as shown in Fig. 12(c). It can be found that the guided waves are well suppressed compared with Fig. 12(a), though the near field of the source and approximation in material parameter extraction introduces some errors compared with the material in Fig. 12(b). Embedded metal mesh could cause more fabrication complexity and costs. An alternative is proposed by printing the metal mesh on the bottom side of the meta-cover. Fig. 12(c) and (d)

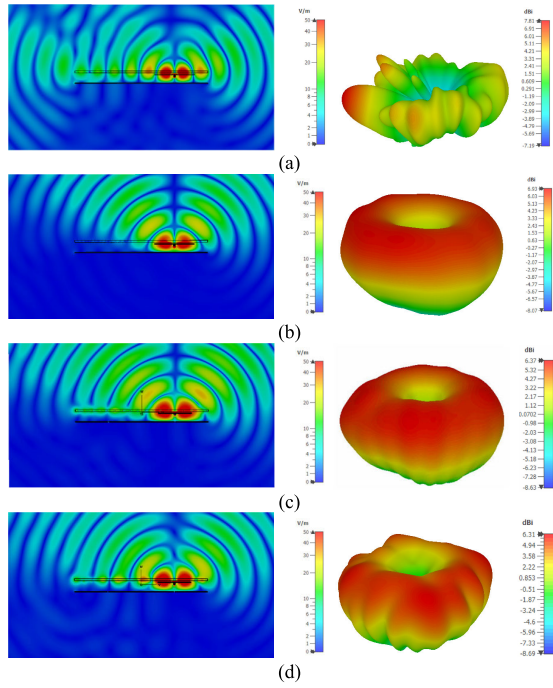


Fig. 12. E -field distribution of an off-center FDR under covers with different materials at $x = 50$ cutting plane and corresponding radiation patterns. The cover are designed with materials with (a) $\epsilon_t = 4$, (b) $\epsilon_t = -0.1$, and meta-cover unit cells with (c) middle-placed metal mesh and (d) bottom-placed metal mesh.

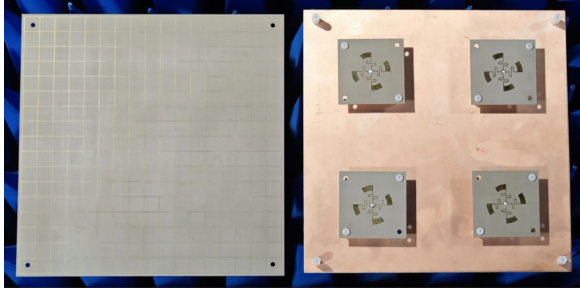


Fig. 13. Prototypes of FDRs (right) with a meta-cover (left).

shows the comparison with an acceptable difference between the corresponding magnitude of the E -field and the radiation pattern. Therefore, the bottom metal mesh unit cells are utilized for fabrication and measurements.

IV. FABRICATION AND EXPERIMENTS

The proposed MIMO antenna systems, consisting of four FDRs sharing a finite ground and covered by a meta-cover, were prototyped and measured, as shown in Fig. 13, where all substrates for the PCBs are F4BTMS450 ($\epsilon_r = 4.5$). The white columns are Industrial Nylon ($\epsilon_r = 4$) for supporting the cover and antennas, which have negligible influence on impedance matching and radiation patterns.

Fig. 14 shows the simulated and measured S -parameters of the antennas under dielectric cover and meta-cover. For clarity, only $|S_{11}|$, $|S_{21}|$, and $|S_{31}|$ are presented due to symmetry and reciprocity. It can be found that both antennas are well-matched over the range from 5 to 6 GHz. Meanwhile, the measured $|S_{21}|$ and $|S_{31}|$ are suppressed from -21 to -29 dB and -21 to -30 dB, respectively, by suppressing the guided waves.

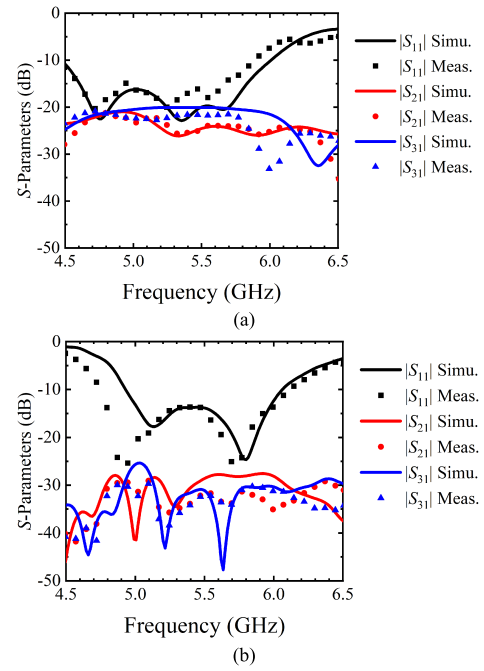


Fig. 14. Simulated and measured S -parameters. FDRs under (a) dielectric cover and (b) meta-cover.

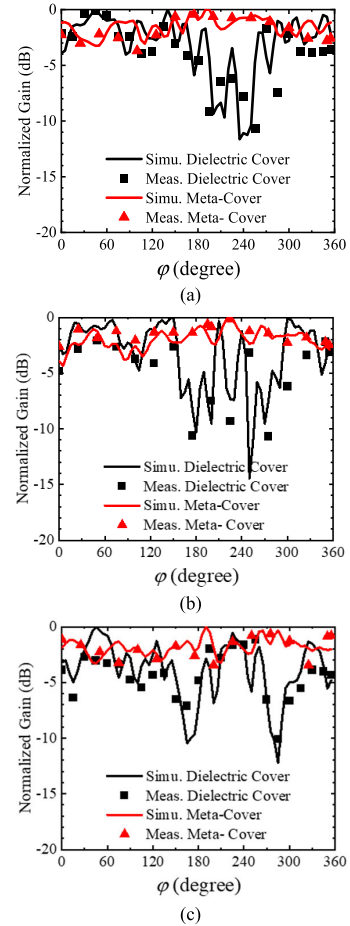


Fig. 15. Simulated and measured radiation patterns at (a) 5, (b) 5.5, and (c) 6 GHz.

The simulated and measured radiation patterns are presented in Fig. 15 at 5, 5.5, and 6 GHz in the $\theta = 60^\circ$ cutting plane for a practical scenario where an AP antenna is mounted on a

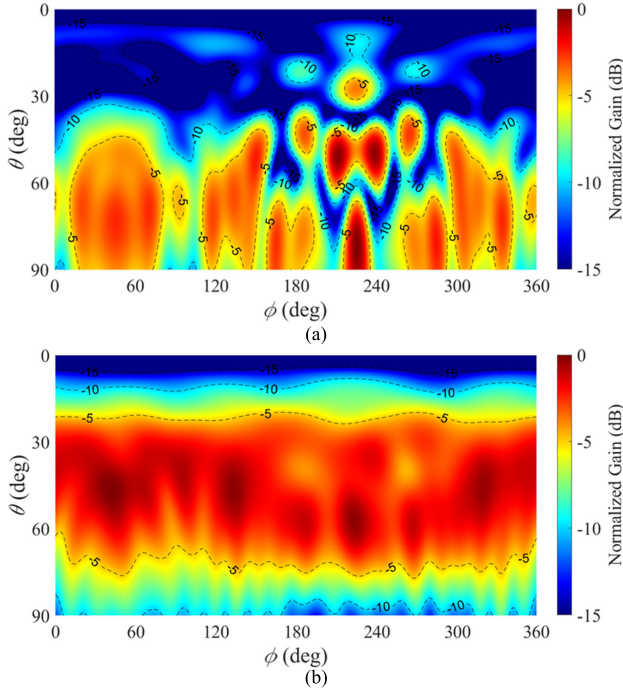


Fig. 16. 2-D radiation patterns at 5.5 GHz. FDRs under (a) dielectric cover and (b) meta-cover.

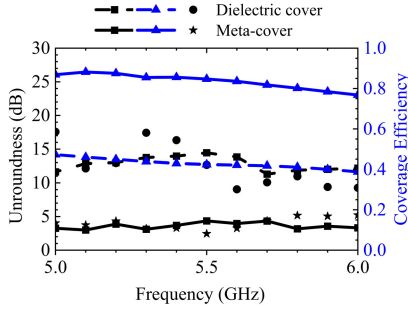


Fig. 17. Unroundness (black) in $\theta = 60^\circ$ cutting plane and coverage efficiency (blue) in $0^\circ < \theta < 75^\circ$ (lines for simulation results and dots for measured results).

ceiling. Omni-directional radiation patterns at larger elevation angles (θ) correspond to longer communication distances as required by industrial standards and discussed in related [1], [2], [3], [4]. The simulated and measured patterns are well agreed upon, and the radiation nulls are significantly alleviated by applying the meta-cover. All the radiation patterns of the antennas with the meta-cover are more omnidirectional.

Furthermore, the 2-D upper-space radiation patterns of antennas under dielectric cover and meta-cover at 5.5 GHz are plotted in Fig. 16. It is found that the radiation pattern of the antenna under meta-cover is more omnidirectional in the whole upper space ($0^\circ < \theta < 90^\circ$), and most energy radiated in space of $\theta < 70^\circ$, ensuring uniform patterns, high coverage efficiency, and low interface with nearby APs or devices. The simulated and measured unroundness in $\theta = 60^\circ$ cutting plane, defined as the maximum gain difference within the cross section of the radiation pattern at a given angle θ , are plotted in Fig. 17. Over the range from 5 to 6 GHz, the measured unroundness is alleviated from 17.6 to 5.2 dB. Meanwhile, the coverage efficiency, the ratio of the radiated power within

$0^\circ < \theta < 75^\circ$ to the total radiated power, is improved from 42.4% to 84.7%.

V. CONCLUSION

The conditions of suppressing guided waves along a uniaxial anisotropic metamaterial slab have been derived and applied to the design of a meta-cover for improving the unroundness of the radiation patterns of off-center covered dipoles. The cutting-off conditions of guided waves have been derived with no prerequisites on the tensor metamaterial elements. After the investigation of pattern distortion of the covered horizontally polarized FDRs, a meta-cover has been designed to suppress guided waves with high transmission properties. The MIMO antennas covered by the proposed meta-cover have improved the radiation patterns for omnidirectional radiation and doubled the coverage efficiency. The proposed method is promising for high capability, stable, and robust communication systems, such as industrial networks, 5G micro base stations, and WiFi 6 APs.

APPENDIX

The comparison between the Lorentz model and the Drude model is presented here for the desire to satisfy the guided wave suppression conditions and wideband properties in the design of the meta-cover. Generally, the Lorentz model of relative permittivity is expressed as [57]

$$\epsilon_{r,\text{Lorentz}}(\omega) = 1 + \frac{\chi_L}{-\omega^2 + j\Gamma_L\omega + \omega_0^2} \quad (18)$$

where χ_L is coupling coefficient, Γ_L is damping coefficient, and characteristic frequency of the restoring forces ω_0 . When the restoring force is negligible, one obtains the Drude model

$$\epsilon_{r,\text{Drude}}(\omega) = 1 + \frac{\chi_D}{-\omega^2 + j\Gamma_D\omega}. \quad (19)$$

Considering that the working conditions of the proposed meta-cover require near zero transverse relative permittivity ϵ_t , the dispersive properties of the two models are discussed at $\text{Re}(\epsilon_r) = 0$ for the desire of wideband design. Generally, the image part of ϵ_r approximately vanishes at the interested band. For simplicity, the damping coefficients of both models are neglected ($\Gamma_L = \Gamma_D = 0$). And when $\epsilon_r = 0$, we have

$$\text{Lorentz Model: } \omega_{\text{Lorentz}}^2|_{\epsilon_r=0} = \chi_L + \omega_0^2 \quad (20)$$

$$\text{Drude Model: } \omega_{\text{Drude}}^2|_{\epsilon_r=0} = \chi_D. \quad (21)$$

Then, differentiating (18) and (19) with respect to ω yields

$$\frac{d(\epsilon_{r,\text{Lorentz}})}{d\omega} = \frac{2\omega\chi_L}{(\omega_0^2 - \omega^2)^2} \quad (22)$$

$$\frac{d(\epsilon_{r,\text{Drude}})}{d\omega} = \frac{2\chi_D}{\omega^3}. \quad (23)$$

Substitute (20) and (21) into (22) and (23), respectively, we obtain

$$\left. \frac{d(\epsilon_{r,\text{Lorentz}})}{d\omega} \right|_{\epsilon_r=0} = \frac{2\omega}{\chi_L} \quad (24)$$

$$\left. \frac{d(\epsilon_{r,\text{Drude}})}{d\omega} \right|_{\epsilon_r=0} = \frac{2}{\omega}. \quad (25)$$

Since $\varepsilon_{r,\text{Lorentz}} = 0$ always appears at a frequency higher than the characteristic frequency ω_0 , i.e., $\omega^2|_{\varepsilon_r=0} - \omega_0^2 > 0$. Thus

$$\left. \frac{d(\varepsilon_{r,\text{Lorentz}})}{d\omega} \right|_{\varepsilon_r=0} = \frac{2\omega}{\chi_L} = \frac{2\omega}{\omega^2 - \omega_0^2} > \frac{2}{\omega} = \left. \frac{d(\varepsilon_{r,\text{Drude}})}{d\omega} \right|_{\varepsilon_r=0}. \quad (26)$$

Therefore, the Drude model is less dispersive than the Lorentz model when $\varepsilon_r = 0$ and presents more potential for wideband design.

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multiinput multioutput antennas.

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